



SeVee Designs

Premium Furniture & Interior Design

System Walkthrough

E-Commerce Platform Documentation

www.seveedesigns.com
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1. Executive Summary

SeVee Designs is a premium Ghanaian furniture and interior design brand offering handcrafted Executive, Office, and Student furniture collections. The platform combines modern e-commerce capabilities with Augmented Reality (AR) technology, allowing customers to visualize furniture in their own spaces before purchasing.

This document provides a comprehensive walkthrough of the SeVee Designs digital platform — covering customer-facing features, the admin management panel, AR technology, payment processing, and the underlying architecture.

2. Platform Overview

What is SeVee Designs?

SeVee Designs is a full-stack e-commerce platform built for a Ghanaian furniture brand. It enables customers to browse products, view them in AR, and purchase online via Paystack (mobile money, card, bank transfer). The platform includes:

- Customer-facing storefront with product catalog
- Augmented Reality (AR) product viewer
- AI-powered chatbot for furniture recommendations
- Admin panel for product, order, and staff management
- Sales reporting and analytics dashboard
- Coupon and promotional tools
- Blog and content management

Key Differentiators

- AR Preview — Customers can see 3D furniture models in their real space
- AI Chatbot — Smart assistant helps customers find the right furniture
- Ghana-Focused — Paystack integration for Mobile Money and local payments
- Handcrafted Quality — Each piece is custom-made with premium materials

3. Technology Stack

Frontend

- React 18 with TypeScript for type-safe development
- Vite for fast builds and hot module replacement
- Tailwind CSS v4 for responsive, utility-first styling
- React Router for client-side navigation
- Axios for API communication
- Recharts for admin analytics charts

Backend

- Express.js (Node.js) RESTful API server
- TypeScript for type safety across the stack
- PostgreSQL database (hosted on Supabase)
- JWT authentication with role-based access control
- Multer for file upload handling
- Google Gemini AI for chatbot intelligence

Infrastructure

- Frontend: Vercel (automatic deployments from GitHub)
- Backend: Render (Node.js hosting with auto-deploy)
- Database: Supabase (PostgreSQL + real-time subscriptions)
- Storage: Supabase Storage (product images CDN)
- Domain: seveedesigns.com

Development Workflow

- ' Code pushed to GitHub !' Auto-deploys to Vercel (frontend) and Render (backend)
- ' Database migrations run via SQL scripts on Supabase dashboard
- ' Environment variables managed per-service on hosting platforms

4. Customer-Facing Features

4.1 Homepage

The homepage serves as the brand's digital storefront, featuring:

- Hero section with brand imagery and call-to-action
- Featured products carousel highlighting top items
- AR showcase section demonstrating the 3D preview capability
- Category navigation (Executive, Office, Student)
- Brand story section
- Footer with contact info, social links, and newsletter signup

[Homepage Screenshot]

Hero section with featured products and AR showcase

4.2 Product Catalog (Shop Page)

The shop page provides a full browsing experience:

- Grid layout showing product cards with images, names, and prices
- Category filtering (Executive, Office, Student collections)
- Search functionality (search by product name or description)
- Sorting options: Price (low/high), Name (A-Z), Newest first
- Pagination for large catalogs
- Wishlist heart icon on each product card

[Shop Page Screenshot]

Product grid with category filters and search bar

4.3 Product Detail Page

Each product has a dedicated detail page showing:

- Product image gallery (multiple images, thumbnail navigation)
- Product name, item code (SKU), and price
- Sale price with original price strikethrough (if on sale)
- Stock availability indicator
- Product specifications (Material, Dimensions, Weight, etc.)
- AR "View in Your Space" button (for products with 3D models)
- Add to Cart button with quantity selector
- Wishlist toggle button
- Related/similar products section

[Product Detail Screenshot]

Image gallery, specs, AR button, and add-to-cart

4.4 Shopping Cart

The cart page manages selected items:

- List of items with images, names, quantities, and prices
- Quantity adjustment (+ / - buttons)
- Remove item from cart
- Subtotal and total calculation
- Checkout button to proceed to payment
- Empty cart state with link back to shop

[Cart Page Screenshot]

Item list with quantities, prices, and checkout button

4.5 Checkout & Payment

The checkout process handles order completion:

- Delivery information form (name, phone, email, address)
- Order summary with itemized costs
- Payment via Paystack (supports Mobile Money, Card, Bank Transfer)
- Secure payment processing (PCI-compliant via Paystack)
- Order confirmation page with order number
- Email confirmation sent after successful payment

[Checkout Page Screenshot]

Delivery form, order summary, and Paystack payment button

4.6 User Account & Dashboard

Registered users have access to:

- Personal profile management (name, email, phone)
- Order history with status tracking
- Wishlist management (save products for later)
- Google OAuth social login (one-click sign-in)
- Secure JWT-based session management

4.7 AI Chatbot

An intelligent assistant available on every page:

- Answers questions about products, materials, and pricing
- Recommends furniture based on customer needs
- Redirects student/school queries to the Student collection
- Powered by Google Gemini AI
- Chat history within session
- Minimized/maximized toggle for non-intrusive browsing

[Chatbot Screenshot]

Chat interface with AI furniture recommendations

5. Admin Panel

The admin panel provides full management capabilities for store owners and staff. Access is restricted to users with "admin" or "superadmin" roles.

5.1 Admin Dashboard

- KPI cards: Total Revenue, Orders, Products, Customers
- Recent orders table with quick status overview
- AI Journal console for business insights
- AR click analytics (how many times AR was used per product)

[Admin Dashboard Screenshot]

KPI cards, recent orders, and analytics

5.2 Product Management

- Product catalog table with search, sort, and filter
- Create new products with name, price, stock, category, description
- Upload product images (multiple images per product)
- Upload 3D models (.glb/.usdz) for AR viewing
- Build product specifications (key-value pairs)
- Toggle product visibility (active/inactive) and featured status
- Edit and delete products

[Admin Products Screenshot]

Product catalog table with edit/delete actions

5.3 Order Management

- View all orders with customer details and items
- Update order status (Pending !' Processing !' Shipped !' Delivered)
- Expandable order details view
- Filter and search orders

5.4 Sales Reports

- Revenue charts (daily, weekly, monthly)
- Product volume analysis
- Regional customer distribution
- Export-ready data views

5.5 Coupon Management

- Create discount coupons (percentage or fixed amount)
- Set expiry dates and usage limits
- Toggle coupon active/inactive status

5.6 Staff Management (Superadmin Only)

- Add new staff accounts with role assignment
- Roles: Admin, Superadmin, Salesperson
- Edit staff profiles and permissions
- Remove staff access

6. Augmented Reality (AR)

The AR feature allows customers to view 3D furniture models in their real-world environment using their smartphone camera. This is a key differentiator for SeVee Designs.

How It Works

1. Admin uploads a 3D model file (.glb or .usdz) when creating/editing a product
2. The 3D model is stored alongside the product in the database
3. Customer opens the product page and taps "View in Your Space"
4. The AR viewer loads the 3D model using the device camera
5. Customer can walk around, resize, and position the furniture in their space
6. Customer can take a screenshot or proceed to purchase

Device Compatibility

- iOS: Safari on iPhone/iPad (uses Quick Look for .usdz files)
- Android: Chrome browser (uses model-viewer for .glb files)
- Desktop: 3D model preview (no AR, but rotatable view)

7. Payment & Order Flow

Payment Gateway: Paystack

SeVee Designs uses Paystack as the payment processor, which is the leading payment gateway in Africa. It supports:

- Mobile Money (MTN, Vodafone, AirtelTigo)
- Debit/Credit Cards (Visa, Mastercard)
- Bank Transfer
- USSD payments

Order Lifecycle

1. Customer adds products to cart
2. Customer fills delivery information at checkout
3. Paystack payment is initiated
4. Payment confirmed !' Order created in database
5. Admin receives notification and processes the order
6. Admin updates status: Processing !' Shipped !' Delivered
7. Customer sees status updates in their dashboard

[Order Flow Diagram]

Visual flow: Cart !' Checkout !' Payment !' Confirmation !' Fulfillment

8. Architecture & Infrastructure

System Architecture

The platform follows a modern three-tier architecture:

Architecture Diagram

- ' Client (Browser) !' Vercel CDN (React SPA)
- ' !•
- ' API Server (Express.js on Render)
- ' !•
- ' Database (PostgreSQL on Supabase)
- ' !•
- ' Storage (Supabase Storage — Product Images)

Deployment Pipeline

- Code is pushed to GitHub repository
- Vercel automatically builds and deploys the frontend
- Render automatically builds and deploys the backend
- Database changes are applied via Supabase SQL editor
- No manual deployment needed — fully automated CI/CD

Data Flow

1. User visits seveedesigns.com !' Vercel serves the React app
2. React app makes API calls to sevee-designs1.onrender.com/api
3. Express server processes requests and queries PostgreSQL
4. Product images are served from Supabase Storage CDN
5. Payments are processed via Paystack API
6. Chatbot queries are sent to Google Gemini AI API

9. Security & Authentication

Authentication

- JWT (JSON Web Tokens) for session management
- Password hashing with bcrypt
- Google OAuth for social login
- Role-based access control (Customer, Salesperson, Admin, Superadmin)

API Security

- CORS configured for allowed origins only
- Rate limiting to prevent abuse
- Helmet.js for HTTP header security
- Input validation on all endpoints
- Admin-only routes protected with middleware

Data Security

- Environment variables for all secrets (never committed to code)
- Supabase Row Level Security (RLS) policies
- HTTPS enforced on all connections
- Paystack handles PCI-compliant card processing

10. Support & Contact

Platform URLs

- Customer Site: <https://seveedesigns.com>
- Admin Panel: <https://seveedesigns.com/admin>
- API Server: <https://sevee-designs1.onrender.com/api>

Key Contacts

- Technical Support: Available via the platform
- GitHub Repository: https://github.com/loverjoy/Sevee_Designs

Maintenance & Updates

- Platform auto-deploys on every GitHub push
- Database backups managed by Supabase
- Server logs available on Render dashboard
- Monitoring via Render built-in metrics

Thank you for choosing SeVee Designs

Premium Furniture. Crafted with Excellence.

www.seveedesigns.com | info@seveedesigns.com